

Stage 1 Tutorial Activity

- Why Software Engineering Still Matters

Stage 1 | Introducing Software Technology Case Study with Python and Guided AI use

Learning goals

- Explain why software engineering is broader than coding.
- Identify stakeholders in a simple software problem.
- Recognise missing requirements.
- Critically evaluate AI-generated feature suggestions.
- Explain why AI output should not automatically be treated as correct.

Activity 1 - Think-Pair-Share (10 minutes)

If ChatGPT or Copilot can produce a 100-line Python application very quickly, what knowledge does a software engineer still need?

1. Code Verification
2. System Analysis and Design
3. Maintenance & Long-Term System Lifecycle

Activity 2 - Is This Software Engineering? (10 minutes)

Scenario A: A student writes a 50-line Python calculator.

Scenario B: A team develops a payroll system used by 5,000 employees.

Scenario C: An AI assistant generates a simple appointment application from one prompt.

Scenario	Programming?	Software engineering?	Why?
A	Yes	No	It is a single-author, has a short lifespan and no real failure impact.
B	No	Yes	It is multi-person maintenance, has a high security/compliance risk and a long lifespan
C	Yes	No	It's a simple app that has no lifecycle management

Activity 3 - SmartCare Problem Analysis (20 minutes)

Client statement: SmartCare Community Clinic currently uses spreadsheets and paper records to manage patients and appointments. The clinic wants new software to improve these processes.

Task 1 - Identify stakeholders

Stakeholder	What do they need?
Receptionists	A streamlined appointment scheduling calendar
Doctors, Nurses, Specialists	Access to patient medical history, clinical notes, and prescription during consultations
Clinic Management / Practice Managers	Reporting / Analytics capabilities and staff scheduling tools
Patients	An easy way to book or reschedule appointments

Task 2 - Identify current problems

1. Risk of Scheduling Errors and Double-Booking
2. Fragmented Patient Histories
3. Data Security Risks
4. Extra administrative work / Inefficient workflows

Task 3 - Ask client questions

1. When receptionists, nurses, and doctors interact with patient charts throughout the day, what are the exact steps taken, where do handoffs happen, and how many staff members need access to the patient's record at peak times?
2. How is patient consent, role-based access, and data retention currently managed across the current paper-based system, and what external Health-related compliance frameworks must the new system align with?
3. What volume of historical paper records exists, how much of it must be migrated to the new platform?
4. What external systems does the new system need to connect with, and what is your current on-premises hardware and network capacity?
5. What is the digital literacy level of the staff, how much downtime can the clinic tolerate during rollout, and who will perform internal training?

Activity 4 - Critique an AI Response (15 minutes)

An AI assistant suggests: appointment management; facial-recognition login; AI diagnosis recommendations; patient search; online payment; practitioner schedule view; insurance processing; automatic treatment-plan generation.

Suggestion	Client evidence?	In scope?	Decision
Appointment management	Supported	Yes	Approved
Facial recognition login	Not supported	No	Rejected
AI diagnosis recommendations	Not supported	No	Rejected
Patient search	Supported	Yes	Approved
Online payment	Weakly Supported	Scope Creep	Deferred (Phase 2)
Practitioner schedule	Supported	Yes	Approved

view			
Insurance processing	Weakly Supported	Scope Creep	Deferred
Treatment-plan generation	Partially Supported	In Scope	Approved

Exit question

Write one activity that a software engineer must perform and that cannot safely be delegated entirely to AI.

Requirements Gathering and Stakeholder Alignment